

Final Project Appendix (code)

Nathaniel Pulsone

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The following is R code that was used to perform Gibbs sampling and analysis for the study.

```
# Load data
setwd("C:/Users/njp09/Documents/Notes/6 - Fall 2025/Bayesian Stats/Final Project")
df = read.csv("diabetes.csv")

# response variable
y = df$DiabetesPedigreeFunction

n = length(y)
p = 7

# predictor variables
preg = df$Pregnancies
gluc = df$Glucose
bp = df$BloodPressure
st = df$SkinThickness
ins = df$Insulin
bmi = df$BMI
age = df$Age

X = cbind(1, preg, gluc, bp, st, ins, bmi, age)
rm(preg, gluc, bp, st, ins, bmi, age)
```

```
# helper functions
SSRgz = function(z){
  Xz = X[,which(z == 1)]
  pz = sum(z)
  result = t(y)%%(diag(1, n) - g/(g+1) * Xz%%solve(t(Xz)%%Xz)%%t(Xz))%y
  return(result)
}

log_py = function(z){
  pz = sum(z)
  SSR = SSRgz(z)

  result =
    -n/2 * log(pi) * lgamma((nu_0 + n)/2) - lgamma(nu_0/2) -
    pz/2 * log(1 + g) + nu_0/2 * log(nu_0*sig2_0) -
    (nu_0 + n)/2 * log(nu_0*sig2_0 + SSR)

  return(result)
}

# priors
g = n
nu_0 = 1
sig2_0 = 5

iters = 1000

# samples
z = rbind(c(1, rbinom(p, 1, 0.5)))
beta = rbind(rep(0, p+1))
sig2 = c()
```

```
# Gibbs Sampling Algorithm
for(s in 1:iters){
  z_curr = as.numeric(z[s,])
  for(j in 2:(p+1)){
    z_in = z_curr
    z_in[j] = 1

    z_out = z_curr
    z_out[j] = 0

    log_oj = log_py(z_in) - log_py(z_out)

    pr = sigmoid(log_oj)

    z_curr[j] = rbinom(1, 1, pr)
  }

  z = rbind(z, z_curr)

  Xz = X[,which(z_curr == 1)]
  SSR = SSRgz(z_curr)
  bhz = solve(t(Xz)%*%Xz)%*%t(Xz)%*%y

  sig2_curr = 1/rgamma(1, (nu_0 + n)/2, (nu_0*sig2_0 + SSR)/2)
  sig2 = c(sig2, sig2_curr)

  Sigma = g/(g+1) * sig2_curr * solve(t(Xz)%*%Xz)
  bz = mvrnorm(1, g/(g+1) * bhz, Sigma)

  beta_curr = c()
  for(j in 1:(p+1)){
    k = 1
    if(z_curr[j] == 1){
      beta_curr = c(beta_curr, bz[k])
      k = k+1
    }
    else{
      beta_curr = c(beta_curr, 0)
    }
  }
  beta = rbind(beta, beta_curr)
}
```